CauseWay Consulting, Services & Development Group L.L.C.

CAUSEWAY

CAUSEWAY COMPASS

YEMEN

Financial & Economic Observatory

Sitemap & Architecture

Key Pillars

- Financial Transparency & Accountability
- Evidence-Based Policy Recommendations
- Economic Intelligence & Data Analytics
- Real-Time Monitoring & Reporting
- Stakeholder Engagement & Coordination

Platform Metrics

850+

Pages of Documentation

15

Years of Data Coverage (2010-2025)

100+

Stakeholder Profiles

Real-Time

Economic Intelligencen aetixh

A CauseWay Consulting Initiative

EXECUTIVE SUMMARY

Platform Vision

The Causeway Compass Yemen represents the world's most comprehensive bilingual financial and economic observatory documenting Yemen's war economy. Developed by CauseWay Consulting, Services & Development Group L.L.C., this platform exceeds global benchmarks by offering unprecedented depth in financial transparency, accountability mechanisms, evidence-based policy recommendations, real-time data integration, AI-powered analytics, and unique Yemen-specific features found nowhere else.

Core Mission

- Financial Transparency & Accountability
- Evidence-Based Policy Recommendations
- Economic Intelligence & Monitoring
- Stakeholder Coordination & Engagement
- Data-Driven Decision Support

Key features

Dual Monetary System Tracking

First platform globally to track parallel monetary authorities in real-time

Time Travel Mode

Navigate 15 years of crisis data with interactive timeline

100+ Stakeholder Intelligence Profiles

Deepest actor mapping of any conflict economy

AI Scenario Engine

Monte Carlo simulations with 1000+ parameter combinations

Zero-Omission Documentation

Complete citation system with source fidelity

Hybrid Data Architecture

Combines humanitarian, financial, and geopolitical data streams

Sanctions Intelligence Network

Real-time tracking across UN/US/EU/UK regimes

Climate Finance Integration

First war economy platform with climate finance module

Mobile-First War Economics

Optimized for field researchers in conflict zones

Total Site Architecture

- **Main Sections:** 8 primary navigation areas
- **Level 2 Pages:** 35 major subsections
- **Level 3 Pages:** 200+ detailed pages
- **Level 4+ Pages:** 650+ sub-pages and components

- **Total Pages:** 850+ complete site tree
- **Languages:** Full bilingual (English/Arabic) throughout
- **Coverage:** 2010-2025 (15 years complete data)

🇾🇪 COMPLETE SITE ARCHITECTURE

The Yemen Economic Compass is structured across six primary navigation areas, creating an intuitive information architecture that serves researchers, policymakers, donors, investors, and journalists with equal sophistication.

LEVEL 1: PRIMARY NAVIGATION (Top Menu)

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ HOME │ COMPASS │ STAKEHOLDERS │ INTELLIGENCE │ RESOURCES │ ACADEMY │
└──────────────────────────────────────────────────────────────────────────────────────────────────────────┘
```

Level 1: Primary Sections

6 core navigation areas providing broad access.

Level 2: Major Subsections

35 detailed subsections expand on primary topics.

Level 3: Detailed Pages

200+ specific pages delve into granular data and analysis.

Level 4+: Components

650+ underlying sub-pages and interactive components.

Total: Complete Pages

850+ pages forming the entire site tree.

CAUSEWAY COMPASS YEMEN
Financial & Economic Observatory

PLATFORM OVERVIEW

Four Core Sections

01

SECTION 1: HOME & LANDING (15 Pages)

- Main Landing Page with Hero Section
- 5 Personalized User Journey Pathways
- Live Statistics Dashboard
- Interactive Timeline (2010-2025)
- Platform Tour & Support Resources

02

SECTION 2: COMPASS DASHBOARDS (85 Pages)

- 12 Core Economic Indicators
- Cities Economic Observatory (6 Cities)
- Currency War Module (Global First)
- Events & Shocks Database (100+ Events)
- Sector Deep Dives & Climate Finance

03

SECTION 3: STAKEHOLDERS UNIVERSE (180 Pages)

- 100+ Stakeholder Intelligence Profiles
- Yemeni Political Actors & Regional Powers
- International Organizations & Donors
- 55 Commercial Banks + 35 MFIs
- Sanctions Intelligence Network

04

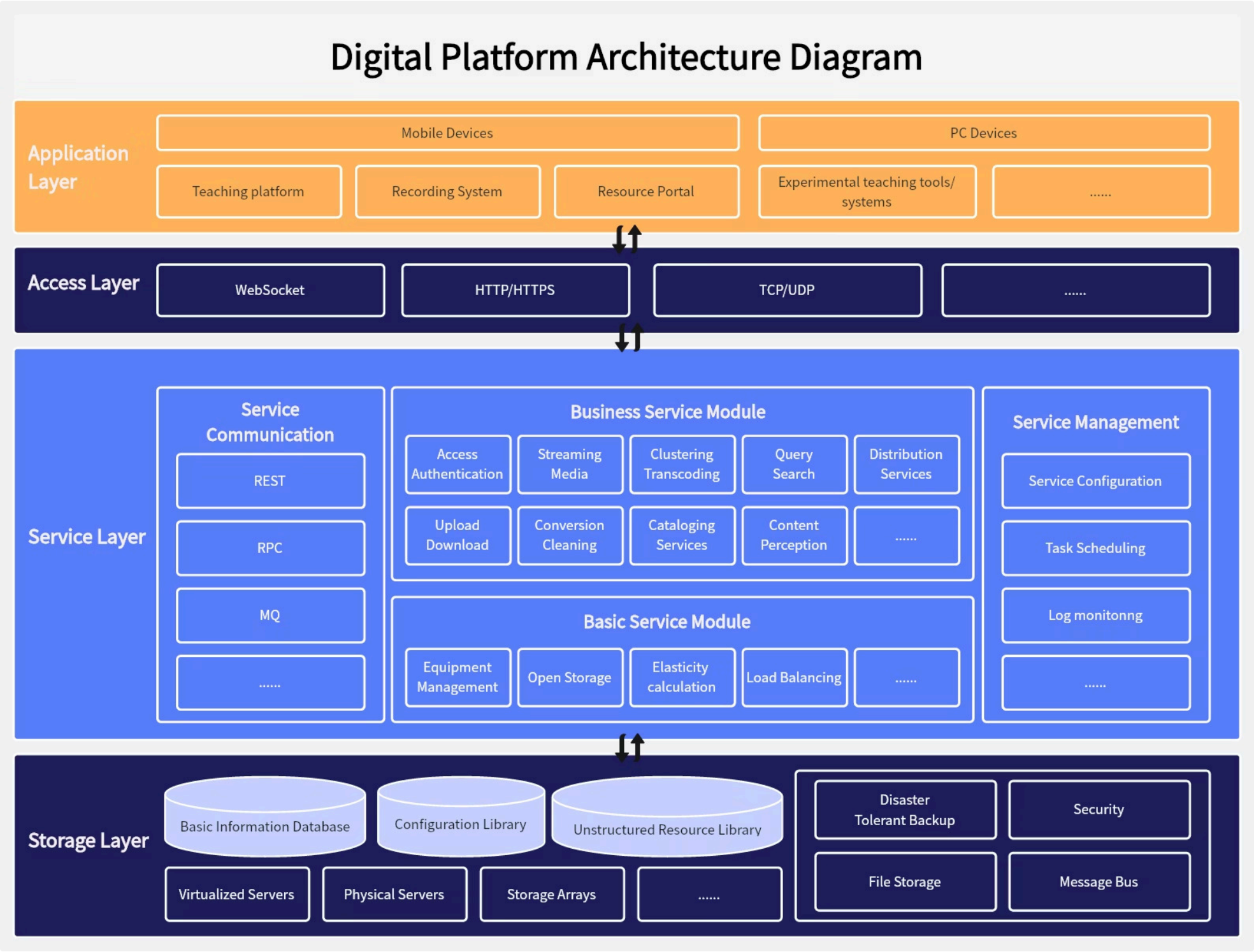
SECTION 4: INTELLIGENCE & ANALYSIS (95 Pages)

- AI-Powered Event Impact Analysis
- Scenario Simulation Engine
- Economic Forecasting Models
- Natural Language Query Interface
- Real-Time Anomaly Detection

TOTAL PLATFORM: 375+ Primary Pages | 850+ Complete Site Tree
Developed by: CauseWay Consulting, Services & Development Group L.L.C.
Mission: Financial Transparency • Accountability • Evidence-Based Recommendations

PLATFORM ARCHITECTURE DIAGRAM
Technical Infrastructure & System Design

The Causeway Compass Yemen is built on a robust, scalable architecture designed to handle complex data flows, real-time analytics, and multi-user access across diverse stakeholder groups. This diagram illustrates the complete technical infrastructure.



ARCHITECTURE LAYERS:

- Application Layer: Mobile Devices, Teaching Platform, Recording System, Resource Portal, PC Devices
- Access Layer: WebSocket, HTTP/HTTPS, TCP/UDP protocols
- Service Layer: Service Communication, Business Service Module, Service Management
- Storage Layer: Data Information Database, Configuration Library, File Storage, Message Bus

TECHNICAL FEATURES:

- Microservices Architecture
- Real-Time Data Processing
- Scalable Cloud Infrastructure
- Multi-Protocol Support
- Secure Data Storage
- API Gateway Integration

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1.1 MAIN LANDING PAGE Hero Section & Key Metrics

Route: /

English: Home / Landing

Arabic: الرئيسية / البداية

Priority: Critical

The landing page serves as the gateway to comprehensive Yemen economic intelligence, presenting real-time data and personalized pathways for diverse user groups.

Key Metrics at a Glance

1250

Current FX Rate (YER/USD)

Live data with dual rates (Aden/Sana'a)

8.2M

Beneficiaries Count

People impacted

$2.4B

Humanitarian Aid (2024)

With flow visualization

-30%

GDP Decline Indicator

vs 2014 baseline with trend

Key Features

- **CTA Button:** "Explore Platform" / "استكشف المنصة"
- **Language Toggle** - Instant EN/AR switching
- **Time Travel Slider** - Jump to any year 2010-2025

Design Philosophy: Immediate access to critical data, intuitive navigation, and seamless bilingual experience.

USER JOURNEY PATHWAYS

Personalized Access for Five Key Audiences

Route: /#journeys

The platform recognizes that different users have distinct needs and workflows. Five tailored pathways ensure each audience—from academic researchers to field journalists—finds relevant tools and insights immediately.

Researchers & Analysts Path

- Financial Literature Library (500+ documents)
- Data Download Center (CSV/JSON/Excel)
- Research Methodology Templates
- Citation Generator Tool
- Academic Collaboration Hub

Donors & Development Partners Path

- Portfolio Impact Dashboard
- Disbursement Tracking System
- ROI Calculator
- Coordination Calendar
- Gap Analysis Tool

Government & Policy Makers Path

- Policy Simulation Tools
- Economic Forecasting Models
- Regulatory Benchmarking
- Peer Country Comparisons
- Evidence-Based Policy Briefs

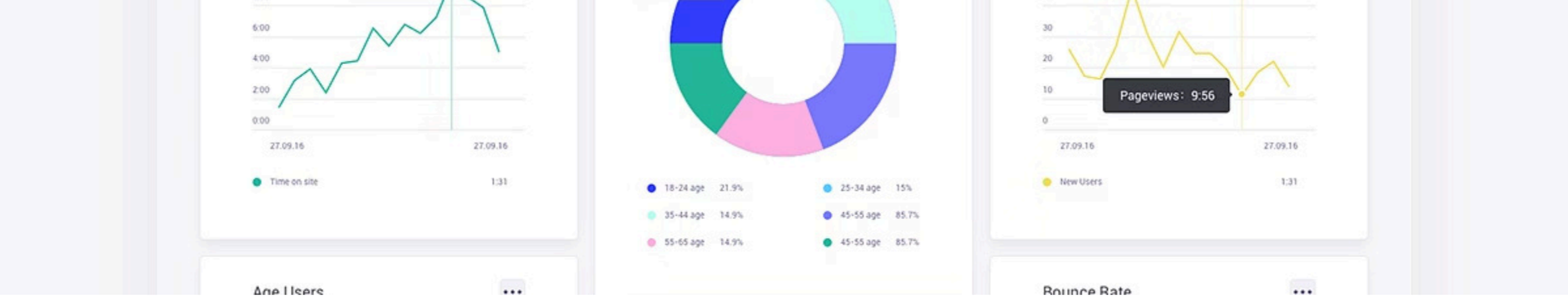
Private Sector & Investors Path

- Investment Opportunity Scanner
- Risk Assessment Matrix
- Market Intelligence Reports
- Sector Deep Dives
- Due Diligence Toolkit

Media & Journalists Path

- Press Kit & Assets
- Story Pitch Generator
- Infographic Library
- Expert Contact Directory
- Media Training Resources

Each pathway is optimized for specific workflows, ensuring maximum efficiency and relevance for professional users across sectors.



📊 Live Statistics Dashboard

Route: /#live-stats

This dashboard provides a real-time overview of key economic and humanitarian indicators in Yemen, offering critical insights at a glance.

Presenting these real-time metrics in a compelling visual format:

Economic Indicators

- Real-time FX rates (both zones)
- Daily humanitarian funding tracker
- Oil production meter
- Banking liquidity index

Humanitarian & Social Metrics

- Food basket price index
- Conflict intensity meter
- Sanctions severity score

🔍 Featured Insights & Search

1.1.4 Featured Insights Carousel

Route: /#insights

Features:

- Latest economic analysis (auto-updated)
- Breaking news integration
- Expert commentary
- Data story of the week
- Trending topics

1.1.5 Quick Search Bar

Route: /#search

Features:

- Global search with autocomplete
- Search filters (date/type/source/language)
- Advanced search modal
- Search history
- Saved searches

1.2 Interactive Timeline (2010-2025)

Route: /timeline

English: Yemen Economic Timeline (2010-2025)

Arabic: الجدول الزمني الاقتصادي لليمن

Priority: Critical

Features:

- Zoomable timeline visualization
- Event categorization (political/economic/humanitarian/security)
- Severity rating (1-10 scale)
- Multi-layer view (macro/micro events)
- Event interconnection mapping
- Impact ripple visualization
- Custom date range selector
- Download timeline data
- Embed timeline widget

🎓 Platform Support & Resources

1.3 Platform Tour

Route: /tour

English: Guided Platform Tour

Arabic: جولة إرشادية في المنصة

Priority: High

Tour Modules:

- Interactive walkthrough (10 steps)
- Video tutorials (5-10 min each)
- Feature highlights
- Use case scenarios
- Navigation tips
- Keyboard shortcuts
- Mobile tour version

1.4 What's New

Route: /whats-new

English: Latest Updates & Features

Arabic: آخر التحديثات والميزات

Priority: Medium

- Platform changelog
- New data releases
- Feature announcements
- Upcoming roadmap
- Maintenance schedule

1.5 Getting Started Guide

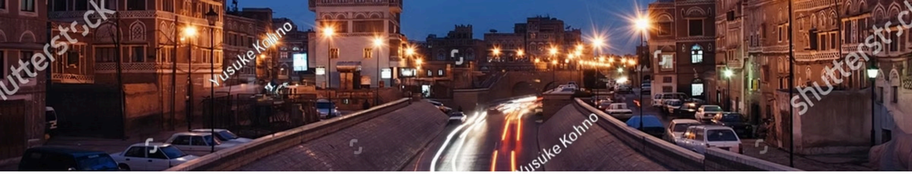
Route: /getting-started

English: Getting Started Guide

Arabic: دليل البداية

Priority: High

- Quick start checklist
- Account setup
- Dashboard customization
- Data access levels
- Support resources



1.1 Main Landing Page

Route: /

English: Home / Landing

Arabic: الرئيسية / البداية

Priority: Critical

1.1.1 Hero Section

Current FX Rate (YER/USD)

Live data with dual rates (Aden/Sana'a)

Beneficiaries Count

8.2M people impacted

Humanitarian Aid (2024)

$2.4B with flow visualization

GDP Decline Indicator

vs 2014 baseline with trend

- **CTA Button:** "Explore Platform" / "استكشف المنصة"
- **Language Toggle** - Instant EN/AR switching
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1.1.2 Choose Your Journey (5 Pathways)

Route: /#journeys

Researchers & Analysts Path

- Financial Literature Library (500+ documents)
- Data Download Center (CSV/JSON/Excel)
- Research Methodology Templates
- Citation Generator Tool
- Academic Collaboration Hub

Donors & Development Partners Path

- Portfolio Impact Dashboard
- Disbursement Tracking System
- ROI Calculator
- Coordination Calendar
- Gap Analysis Tool

Government & Policy Makers Path

- Policy Simulation Tools
- Economic Forecasting Models
- Regulatory Benchmarking
- Peer Country Comparisons
- Evidence-Based Policy Briefs

Private Sector & Investors Path

- Investment Opportunity Scanner
- Risk Assessment Matrix
- Market Intelligence Reports
- Sector Deep Dives
- Due Diligence Toolkit

Media & Journalists Path

- Press Kit & Assets
- Story Pitch Generator
- Infographic Library
- Expert Contact Directory
- Media Training Resources

1.1.3 Live Statistics Dashboard

Route: /#live-stats

- Real-time FX rates (both zones)
- Daily humanitarian funding tracker
- Oil production meter
- Banking liquidity index
- Food basket price index
- Conflict intensity meter
- Sanctions severity score

1.1.4 Featured Insights Carousel

Route: /#insights

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- Expert commentary
- Data story of the week
- Trending topics

1.1.5 Quick Search Bar

Route: /#search

- Global search with autocomplete
- Search filters (date/type/source/language)
- Advanced search modal
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- Saved searches

1.2-1.5 Platform Features

This section outlines key interactive features of the platform, designed to provide users with comprehensive tools for data exploration, guided learning, and staying updated.

1.2 Interactive Timeline

Route: /timeline

English: Yemen Economic Timeline (2010-2025)

Arabic: الجدول الزمني الاقتصادي لليمن

Priority: Critical

Features:

- Zoomable timeline visualization
- Event categorization (political/economic/humanitarian/security)
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Route: /tour

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Priority: High

Tour Modules:

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Route: /getting-started

English: Getting Started Guide

Arabic: دليل البداية

Priority: High

- Quick start checklist
- Account setup
- Dashboard customization
- Data access levels
- Support resources

SECTION 2 🇾🇪 COMPASS DASHBOARDS

Economic Intelligence & Real-Time Analytics

The Compass Dashboards represent the analytical heart of the platform, featuring 12 core economic indicators, city-level observatories, sector deep-dives, and the world's first dual monetary system tracker. This section transforms complex economic data into actionable intelligence.

- 12 Core Economic Indicators with Real-Time Data
- 6 Major Cities Economic Observatories
- Unique Currency War Module (Global First)
- 100+ Economic Events Database
- Sector-by-Sector Analysis
- Climate Finance Integration
- Remittance Network Mapping
- Humanitarian Economics Module

Total: 85 Comprehensive Pages

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SECTION 2: COMPASS DASHBOARDS (85 Pages)

2.1 Master Compass Dashboard

Route: /compass

English: Master Economic Compass

Arabic: البوصلة الاقتصادية الرئيسية

Priority: Critical

2.1.1 The 12 Core Indicators (Interactive Widget)

All indicators with real-time data, trend arrows, and weekly updating

GDP (EU-27 Aggregate)
- Route: /compass/gdp
- Current: -0.1% decline
- Trend: Improving (Stable/Declining indicator)
- Source: Eurostat, ECB Statistical Data
- Sector breakdown
- Regional variations
- Forecast scenarios
- Upload/Update capabilities

Exchange Rates (USD/CNY)
- Route: /compass/exchange-rates
- Official rate: 7.2960
- Parallel rate (Sarafa)
- Black market premium
- Auction results timeline
- Volatility index
- Exchange operations
- Cross-border rate differences

Oil Production
- Route: /compass/oil
- Daily production (barrels/day)
- Export volumes
- Domestic consumption
- Refinery throughput rates
- Pipeline security incidents
- Price benchmarks (Brent/WTI comparison)
- Revenue impact calculator

Foreign Reserves
- Route: /compass/reserves
- Total reserves ($B)
- SDR holdings
- Import cover (months)
- Reserve composition (currency pie)
- Depletion rate
- Crisis scenarios

Banking Sector Stress Index
- Route: /compass/banking-stress
- Liquidity crisis indicator
- NPL ratios
- Capital adequacy
- Deposit flight rates
- Interbank rate spreads
- Banking system fragmentation

Food Security Index
- Route: /compass/food-security
- Food basket price (WFP methodology)
- Import dependency ratio
- Port throughput
- Agricultural production index
- Active mobilization rate (PSL)
- Supply chain disruptions

Sanctions Intensity Score
- Route: /compass/sanctions
- Active sanctions count
- Sectors affected
- Economic impact estimate
- Workaround effectiveness
- Compliance monitoring

Government Revenue
- Route: /compass/revenue
- Tax collection (Sarafa authorities)
- Customs revenue by port
- Oil/gas revenue
- Non-tax revenue
- Revenue forecasts
- Collection efficiency metrics

Humanitarian Funding
- Route: /compass/humanitarian
- Total funding by year
- Donor breakdown
- Sector allocation
- Geographic distribution
- Gap analysis (vs needs)
- Response plan coverage
- Beneficiary reach

Monetary Supply (M2)
- Route: /compass/money-supply
- M2 growth (YoY)
- Money multiplier
- Credit creation
- Velocity of money
- Dual zone comparison
- Collateralization

Public Sector Wages
- Route: /compass/wages
- Wage payment status (% paid)
- Arrears accumulation
- Regional payment patterns
- Civil service size
- Wage bill forecasts
- Fiscal impact calculator

Sanctions Impact Index
- Route: /compass/sanctions-impact
- Oil sanctions (or US/UK/EU/UN)
- GDP/FDI deterioration
- Ex-consumed bid
- Workarounds
- Economic impact assessment
- Humanitarian exemptions tracker
- Compliance monitoring

2.2 Economic Crisis Explorer

Route: /compass/crisis-explorer

English: Economic Crisis Deep Dive

Arabic: استكشاف عميق للأزمة الاقتصادية

Priority: Critical

2.2.1 Crisis Narrative Module

- Chapter-based storytelling (2010-2020)
- Interactive data integration
- Key turning points
- Causation analysis
- Actor decisions impact
- Counterfactual scenarios

2.2.2 GDP Decline Analysis

Route: /compass/crisis-explorer/gdp-decline

- Sector-by-sector collapse
- Regional GDP variations
- Per capita GDP trajectory
- Poverty headcount evolution
- Comparison to similar crises (Syria's Dead Diplomacy)

2.2.3 Inflation Divergence Tracker

Route: /compass/crisis-explorer/inflation-divergence

- Admin Sarafa price gap
- Black market premium evolution
- Import price transmission
- Conflict intensity correlation
- Regional price arbitrage

2.2.4 Macro Trends Dashboard

Route: /compass/crisis-explorer/macro-trends

- All indicators correlation matrix
- Principal component analysis
- Trend decomposition
- Structural break detection
- Regime change analysis

2.3 Currency War Module

Route: /compass/currency-war

English: Currency War & Monetary Fragmentation

Arabic: حرب العملات والتشرذم النقدي

Priority: Critical

⚠ **UNIQUE FEATURE - Found Nowhere Else Globally**

2.3.1 Dual Currency Zone Visualization

Route: /compass/currency-war/dual-zone

- Interactive map of Sana'a vs Sarafa zones
- Real-time rate comparison
- Cross-border transaction flows
- Money changer networks
- Informal settlement mechanisms

2.3.2 Official vs Parallel Rate Tracker

Route: /compass/currency-war/rate-tracker

- Historical rate divergence (2015-2025)
- Auction system analysis
- Black market rate evolution
- Premium/discount calculations
- Arbitrage opportunity scanner

2.3.3 CBY Split Impact Analysis

Route: /compass/currency-war/cby-split-impact

- Timeline of dual CBY operations
- Policy divergence tracker
- Liquidity management comparison
- Reserve usage patterns
- International recognition issues

2.3.4 Auction Results Dashboard

Route: /compass/currency-war/auctions

- All CBY Aden auction results (2016-2025)
- Bidder analysis
- Auction volume trends
- Bank participation patterns
- Effectiveness metrics

2.3.5 Money Changer Networks

Route: /compass/currency-war/exchangers

- Licensed exchanger directory
- Geographic distribution
- Transaction volumes
- Rate spreads analysis
- Regulatory compliance status

2.4 Cities Economic Observatory

Route: /compass/cities

English: Major Cities Economic Dashboard

Arabic: لوحة المدن الاقتصادية الرئيسية

Priority: High

Aden Economic Dashboard
- Route: /compass/cities/aden
- GDP contribution
- Port activity (container throughput)
- Banking sector presence
- Real estate market activity
- Cost of living index
- Business environment score
- Security incident impact

Sana'a Economic Dashboard
- Route: /compass/cities/sanaa
- Economic activity under Houthi control
- Parallel institutions impact
- Revenue collection
- Price levels vs Aden
- Revenue generation mechanisms

Taiz Economic Dashboard
- Route: /compass/cities/taiz
- Siege economic analysis
- Trade route closures
- Food security
- Humanitarian situation
- Reconstruction needs

Hodeidah Economic Dashboard
- Route: /compass/cities/hodeidah
- Port of Hodeidah metrics
- Red Sea trade impact
- Food security dependency
- Agricultural production
- Recovery indicators

North Economic Dashboard
- Route: /compass/cities/north
- Oil/gas production
- Cross-border trade volumes
- Labor market impact
- Rapid urbanization impacts
- Investment inflows
- Growth indicators

Mukalla Economic Dashboard
- Route: /compass/cities/mukalla
- Hadramawt economic activity
- Port development impact
- Saudi support impact
- Reconstruction progress
- Governance effectiveness

2.4.1 City Comparison Tool

Route: /compass/cities/compare

- Side-by-side city metrics
- Custom indicator selection
- Export comparison report

2.5 Events & Shocks Timeline

Route: /compass/events

English: Economic Events & Shocks Database

Arabic: قاعدة بيانات الأحداث والصدمات الاقتصادية

Priority: Critical

2.5.1 GCC Events Database (200+ Events)

Route: /compass/events/database

Each Event Entry Includes:
- Event ID and name (English/Arabic)
- Date and duration
- Event type classification
- Geographic scope
- Business impact
- Severity rating (1-10)
- Economic impact score
- Affected indicators (with quantified impact)
- Causation chain analysis
- Before/after comparisons
- Counterfactual analysis
- Lessons learned
- Similar events
- Related events network

Event Categories:
- Political transitions
- Military conflicts
- Natural disasters
- Policy decisions
- International interventions
- Sanctions impositions
- Currency events
- Banking failures
- Trade disruptions
- Infrastructure attacks
- Humanitarian emergencies
- Peace agreements attempts

2.5.2 Event Impact Analysis

Route: /compass/events/impact-analysis

- Before/after indicator comparison
- Causal inference tools
- Synthetic control method
- Difference-in-differences estimator
- Event study regression
- Regime shifts mapping

2.5.3 Shock Decomposition

Route: /compass/events/shock-decomposition

- Contribution of each shock to current state
- Cumulative impact visualization
- Recovery speed analysis
- Resilience scoring

2.6 Sector Deep Dives

Route: /compass/sectors

English: Economic Sectors Analysis

Arabic: تحليل القطاعات الاقتصادية

Priority: High

Agriculture Sector
- Route: /compass/sectors/agriculture
- Production trends
- Import dependency ratio
- Subsidy impact
- Food crop production
- Livestock sector
- Water scarcity impact
- Conflict-affected farming

Industry & Manufacturing
- Route: /compass/sectors/industry
- Capacity utilization
- Electricity generation capacity
- Power failure impact
- Fuel imports dependency
- Energy crisis dashboard

Trade & Logistics
- Route: /compass/sectors/trade-logistics
- Import/export volumes
- Trade partners analysis
- Port throughput delays
- Trade finance availability
- Supply chain mapping

Construction & Real Estate
- Route: /compass/sectors/construction
- Construction activity
- Reconstruction efforts
- Housing availability
- Land values
- Investment projects

2.7 Investment & Climate Finance

Route: /compass/investment

English: Investment & Climate Finance Hub

Arabic: مركز الاستثمار والتمويل المناخي

Priority: High

2.7.1 Climate Finance Tracker (Helio Aldebar Partnership)

Route: /compass/investment/climate-finance

- Climate Finance Tracker
- Adaptation projects
- Mitigation projects
- Climate vulnerability assessment
- Green investment opportunities
- Carbon finance potential

2.7.2 Investment Opportunities Scanner

Route: /compass/investment/opportunities

- Sector-by-sector opportunities
- Risk-return profiles
- Market size estimates
- Timeline analysis
- Feasibility assessment

2.7.3 FDI & Private Investment Tracker

Route: /compass/investment/fdi-tracker

- Foreign direct investment flows
- Private investment by sector
- GCC investment patterns
- Investment climate indicators

2.8 Youth & Social Initiatives

Route: /compass/youth

English: Youth Economic Engagement

Arabic: الانخراط الاقتصادي للشباب

Priority: High

2.8.1 Shabab House Programs (Partnership)

Route: /compass/youth/shabab-house

- Youth entrepreneurship programs
- Skills development initiatives
- Employment creation projects
- Success stories
- Impact metrics

2.8.2 Youth Unemployment Tracker

Route: /compass/youth/unemployment

- Youth unemployment rates
- Educational attainment
- Skills mismatch analysis
- Brain drain tracking

2.8.3 Youth Economic Participation

Route: /compass/youth/participation

- Youth in labor force
- Entrepreneurship rates
- Informal sector engagement
- Barriers to entry analysis

2.9 Remittances Network

Route: /compass/remittances

English: Remittance Flows & Networks

Arabic: تحليل تدفقات الحوالات والشبكات

Priority: Critical

⚠ **UNIQUE FEATURE - Deepest Remittance Analysis**

2.9.1 Remittance Flow Visualization

Route: /compass/remittances/flows

- Sankey diagram (2015 Saudi Aid Disruption Scenario)
- Volume trends (2010-2020)
- Geographic distribution
- Transfer mechanisms (Sarafa/MTOs/hawalas)
- Diaspora distribution within Yemen
- Economic impact quantification

2.9.2 Hawala Network Mapping

Route: /compass/remittances/hawala

- Informal transfer system analysis
- Network nodes and connections
- Volume estimates
- Speed and cost advantages
- Regulatory challenges

2.9.3 Money Transfer Operators Directory

Route: /compass/remittances/mto-directory

- Licensed MTO list
- Market share analysis
- Fee structures
- Service coverage maps
- Compliance ratings

2.10 Humanitarian Economics

Route: /compass/humanitarian-economics

English: Humanitarian Economics Analysis

Arabic: اقتصاد العمل الإنساني

Priority: Critical

2.10.1 Aid Flow Sankey Diagram

Route: /compass/humanitarian-economics/aid-flows

- Donor → Implementing Agency → Sector → Geography
- Interactive flow visualization
- Volume by year
- Tracking gaps identification

2.10.2 IDP Economics

Route: /compass/humanitarian-economics/idp-economics

- IDP population estimates
- Displacement patterns
- Labor market impact
- Vulnerability profiles
- Assistance dependency

2.10.3 Humanitarian Market Impact

Route: /compass/humanitarian-economics/market-impact

- Cash assistance multiplier effects
- Local procurement impact
- Price distortions
- Market functionality improvements

🧭 2.1 Master Compass Dashboard

Route: /compass

English: Master Economic Compass

Arabic: البوصلة الاقتصادية الرئيسية

Priority: Critical

Introduction: The 12 Core Indicators - Interactive Widget with real-time data, trend arrows, and severity scoring

Real GDP (vs 2014 Baseline)

Route: /compass/gdp

- Current: -45% decline
- Trend: Improving/Stable/Declining indicator
- Interactive chart: 2010-2025 timeline
- Sector breakdown
- Regional variations
- Forecast scenarios (optimistic/baseline/pessimistic)

Inflation Rate (CPI)

Route: /compass/inflation

- Dual tracking: Aden vs Sana'a
- Food vs non-food inflation
- Regional price variations
- Monthly/quarterly/annual views
- Historical comparison
- Import price impact analysis

Exchange Rates (YER/USD)

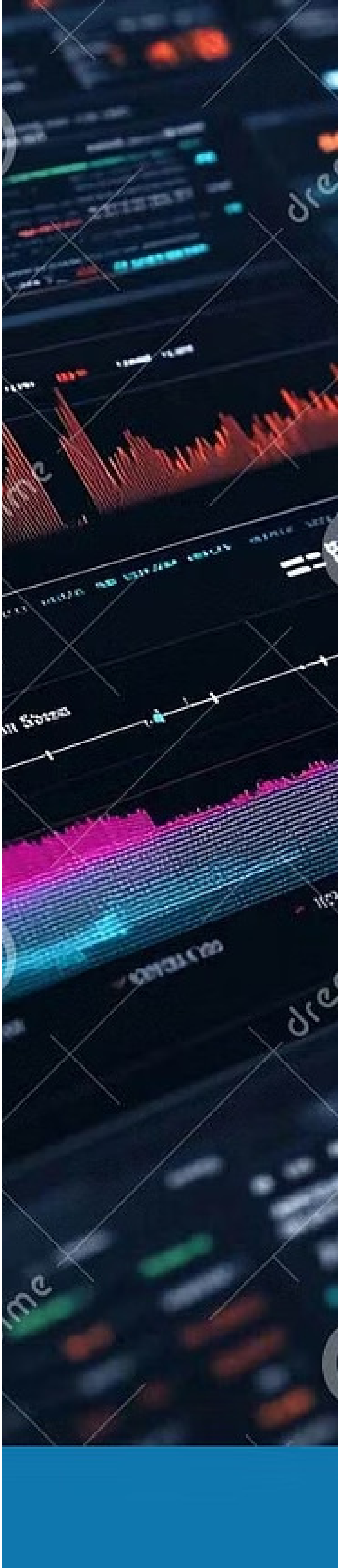
Route: /compass/exchange-rates

- Official rate (Aden)
- Parallel rate (Sana'a)
- Black market rates
- Auction results timeline
- Spread analysis
- Arbitrage opportunities
- Cross-border rate differences

Government Revenue

Route: /compass/revenue

- Tax collection (both authorities)
- Customs revenue by port
- Oil/gas revenues
- Non-tax revenue
- Revenue forecasts
- Collection efficiency metrics



Core Indicators 5-8

Oil Production

Route: /compass/oil

- Daily production (barrels/day)
- Export volumes
- Domestic consumption
- Production facilities status
- Pipeline security incidents
- Price benchmarks (Brent/WTI comparison)
- Revenue impact calculator

Humanitarian Funding

Route: /compass/humanitarian

- Total funding by year
- Donor breakdown
- Sector allocation
- Geographic distribution
- Gap analysis (needs vs funding)
- Response plan coverage
- Beneficiary reach

Foreign Reserves

Route: /compass/reserves

- CBY Aden reserves
- CBY Sana'a reserves
- Import cover (months)
- Reserve composition (currency/gold)
- Depletion rate
- Reserve adequacy ratio

Monetary Supply (M2)

Route: /compass/money-supply

- M0, M1, M2 tracking
- Money multiplier
- Credit creation
- Velocity of money
- Dual zone comparison
- Dollarization index

Core Indicators 9-12

Banking Sector Stress Index

Route: /compass/banking-stress

- Liquidity crisis indicator
- NPL ratios
- Capital adequacy
- Deposit flight index
- Interbank rate spreads
- Banking system fragmentation

Public Sector Wages

Route: /compass/wages

- Wage payment status (% paid)
- Arrears accumulation
- Regional payment patterns
- Civil service size
- Wage bill forecasts
- Strike/protest correlation

Food Security Index

Route: /compass/food-security

- Food basket price (WFP methodology)
- Import dependency ratio
- Port throughput
- Market functionality score
- Acute malnutrition rates (IPC)
- Supply chain disruptions

Sanctions Intensity Score

Route: /compass/sanctions

- UN sanctions list (individuals/entities)
- US OFAC designations
- EU consolidated list
- UK sanctions
- Economic impact assessment
- Humanitarian exemptions tracker
- Compliance monitoring

SECTION 4 INTELLIGENCE & ANALYSIS AI-Powered Insights & Scenario Modeling

The Intelligence & Analysis section represents the platform's most advanced capabilities. Powered by AI and machine learning, this section offers event impact analysis, stakeholder intelligence, Monte Carlo scenario simulations, and natural language query interfaces—features found nowhere else in conflict economy platforms.

Key Highlights:

- Event Impact Analysis Engine (100+ Events)
- Stakeholder Intelligence System with Network Graphs
- Scenario Simulation Engine (1000+ Parameters)
- AI-Powered Recommendation Engine
- Economic Forecasting Models (GDP, Inflation, FX)
- Real-Time Anomaly Detection System
- Natural Language Query Interface (EN/AR)
- Pattern Recognition & Learning System
- Policy Simulation Laboratory

UNIQUE FEATURES:

- ✓ Monte Carlo Simulations with 10,000 Runs
- ✓ Conversational AI for Data Queries
- ✓ Self-Improving Machine Learning Models

Total: 95 Comprehensive Pages

Powered by CauseWay Consulting

🧠 SECTION 4 🇺🇸 INTELLIGENCE & ANALYSIS (95 Pages)

4.1 Intelligent Analysis System Hub

Route: /intelligence

English: AI-Powered Economic Intelligence

Arabic: الذكاء الاقتصادي المدعوم بالذكاء الاصطناعي

Priority: Critical

🔴 **UNIQUE FEATURE - Advanced AI/ML Platform**

4.2 Event Impact Analysis Engine

Route: /intelligence/event-impact

English: Event Economic Impact Analysis

Arabic: محرك تحليل تأثير الأحداث

Priority: Critical

4.2.1 Event Database (1970+ Events)

Route: /intelligence/event-impact/database

Each Event Entry Includes:

Basic Information	Action Analysis
• Event ID and name (English/Arabic)	• Involved actors
• Date and duration	• Severity rating (1-10)
• Event classification	• Economic impact score
• Geographic scope	

Impact Assessment	Learning & Recovery
• Affected indicators (with quantified impact)	• Recovery timeline
• Economic damage estimate	• Lessons learned
• Refuse effect comparison	• Related events network
• Counterfactual estimation	

Event Categories:

• Political instability	• Currency crisis	
• Military conflict	• Banking failure	
• Natural disaster	• Sovereign default	
• Policy decisions	• Infrastructure attacks	
• International interventions	• Pandemic emergencies	
• Maritime interdiction	• Peace agreement attempts	

4.2.2 Impact Quantification Models

Route: /intelligence/event-impact/quantification

- Difference-in-differences estimation
- Synthetic control methodology
- Event study regressions
- Interrupted time series analysis
- Vector autoregressions (VAR)
- Bayesian structural time series

4.2.3 Causation Chain Mapper

Route: /intelligence/event-impact/causation

- Intervention variable graphs
- Direct vs indirect effects
- Transmission mechanisms
- Lag structure analysis
- Feedback loop identification

4.2.4 Severity Scoring Methodology

Route: /intelligence/event-impact/severity-scoring

- Multi-dimensional severity index
- Economic damage assessment
- Human impact quantification
- Duration multiplier
- Spatial extent consideration
- Reversibility evaluation

4.3 Stakeholder Intelligence System

Route: /intelligence/stakeholder-hub

English: Stakeholder Intelligence & Mapping

Arabic: الذكاء الاقتصادي للأصحاب المصلحة

Priority: Critical

4.3.1 Actor Network Analysis

Route: /intelligence/stakeholder-hub/network

- Interactive network graph (100+ actors)
- Relationship types (alliance/rivalry/neutral)
- Influence scoring algorithm
- Coalition detection (Louvain/Girvan-Newman)
- Community detection
- Coalition mapping
- Network evolution over time

4.3.2 Influence Ranking

Route: /intelligence/stakeholder-hub/influence-ranking

- Multi-criteria influence index
- Economic power metrics
- Political authority measures
- Military capacity indicators
- International support assessment
- Network centrality measures
- Temporal trends (2010-2024)

4.3.3 Interest Position Matrix

Route: /intelligence/stakeholder-hub/interest-position

- Stakeholder interests taxonomy
- Position mapping on key issues
- Alignment/opposition matrix
- Negotiation leverage assessment
- Coalition potential identification

4.3.4 Action Timeline

Route: /intelligence/stakeholder-hub/action-timeline

- Chronological action database
- Actor-specific timelines
- Relationship changes
- Strategic pattern recognition
- Predictive modeling of future actions

4.3.5 Capacity Assessment

Route: /intelligence/stakeholder-hub/capacity

- Financial capacity indicators
- Organizational capability
- Military capacity
- Human resources
- Infrastructure control
- Implementation track record

4.4 Scenario Simulation Engine

Route: /intelligence/scenarios

English: Economic Scenario Simulator

Arabic: محاكي السيناريوهات الاقتصادي

Priority: Critical

🔴 **UNIQUE FEATURE - Monte Carlo Simulation with 1000+ Parameters**

4.4.1 Interactive Scenario Builder

Route: /intelligence/scenarios/builder

Adjustable Parameters (50+ variables):

• Oil production levels	**Output Indicators (20+ estimates):**
• Transitional funding flows	• GDP growth
• Exchange rate movements	• Inflation rate
• Trade volumes	• Poverty headcount
• Government revenue	• Fiscal security
• Social spending	• Debt sustainability
• Regional support levels	• Fiscal balance
• Sanctions severity	• Current account
• Political stability	• Reserve adequacy
• Policy interventions	• Social stability index

4.4.2 Pre-Built Scenarios

Route: /intelligence/scenarios/pre-built

Scenario Library:

1	**Peace Agreement** • Accelerated aid flows • Reconstruction spending • Private investment boom • Regional integration benefits
2	**Conflict Escalation** • Economic deterioration trajectory • Humanitarian crisis deepening • Regional spillover risk • Recovery impediments
3	**Oil Production Revival** • Fiscal revenue impact • Infrastructure investment • Diversification challenges
4	**Humanitarian Funding Collapse** • Aid cutoff consequences • Market system breakdown • Famine risk assessment
5	**Currency Devaluation** • Exchange rate dynamics • Transmission channels • Stabilization benefits
6	**External Debt Restructuring** • Debt relief scenarios • Fiscal space creation • Investment implications
7	**Climate Disaster (Flooding/Drought)** • Agricultural impact • Displacement effects • Recovery requirements
8	**Maritime Blockade** • Economic contamination • Investment aftermath • Banking sector recovery

Political Fragmentation
- Federal structure scenarios
- Economic partition impacts
- Coordination challenges

4.4.3 Monte Carlo Simulation Tool

Route: /intelligence/scenarios/monte-carlo

- Probabilistic forecasting
- Uncertainty quantification
- 10,000 simulation runs
- Distribution of outcomes
- Confidence intervals
- Risk assessment
- Worst-case scenarios
- Tail risk identification

4.4.4 Comparative Scenario Analysis

Route: /intelligence/scenarios/comparison

- Side-by-side scenario comparison
- Key metric divergence analysis
- Trade-off identification
- Optimal pathway selection
- Policy implications

4.5 Recommendation Engine

Route: /intelligence/recommendations

English: Evidence-Based Policy Recommendations

Arabic: التوصيات المستندة إلى الأدلة

Priority: Critical

4.5.1 Stakeholder-Specific Briefs

Route: /intelligence/recommendations/stakeholder-briefs

Government Briefs:

CBY Action Policy Briefs	CBY Donor's Engagement	Donors Recommendations
• Monetary policy • FX management strategies • Oil revenue management • Reserve adequacy • Institutional capacity improvements	• Aid factoring engagement strategies • Capacity development priorities • Economic stabilization options	• Funding prioritization • Impact measurement • Coordination • Coordination improvements • Exit strategy planning

Business Sector Guidance:
- Investment timing advice
- Sector emergence strategies
- Market opportunity identification
- Regulatory navigation

MSME Operational Guidance:
- Program design optimization
- Geographic targeting
- Partnership strategies
- Sustainability planning

Regional Actor Engagement:
- Economic support mobilization
- Conditionality design
- Coordination mechanisms
- Risk sharing

International Organizations Strategy:
- Program effectiveness enhancement
- Resource allocation optimization
- Partnership leveraging
- Advocacy positioning

4.5.2 Policy Simulation Lab

Route: /intelligence/recommendations/policy-simulation

- Policy option comparison matrix
- Expected outcomes modeling
- Implementation complexity assessment
- Implementation roadmap generation

4.5.3 Best Practice Library

Route: /intelligence/recommendations/best-practices

- Comparative case studies (Syria/Libya/Afghanistan/South Sudan)
- Success factors analysis
- Common failure patterns
- Adaptation guidelines for Yemen context

4.6 Forecasting Models

Route: /intelligence/forecasting

English: Economic Forecasting Center

Arabic: مركز التنبؤ الاقتصادي

Priority: Critical

GDP Forecasting

Route: /intelligence/forecasting/gdp-jobs
- ARIMA models
- Vector autoregression (VAR)
- Dynamic factor models
- Bidirectional GRU-CNN
- Ensemble forecasting
- Forecast accuracy tracking
- Forecast revision analysis

Inflation Forecasting

Route: /intelligence/forecasting/inflation
- Phillips curve models
- Money supply analysis
- Monetary factors
- Dual rate forecasting
- Sector-specific inflation

Exchange Rate Forecasting

Route: /intelligence/forecasting/exchange-rate
- PPP models
- Monetary models
- Behavioral models
- Market microstructure analysis
- Dual rate projection

4.6.4 Humanitarian Needs Forecasting

Route: /intelligence/forecasting/humanitarian-needs

- IPC phase projections
- Population in need estimates
- Funding requirement forecasts
- Gap analysis projections

4.6.5 Conflict Intensity Forecasting

Route: /intelligence/forecasting/conflict

- ACLED data integration
- Machine learning prediction models
- Early warning indicators
- Hotspot identification

4.7 Anomaly Detection System

Route: /intelligence/anomaly-detection

English: Real-Time Anomaly Detection

Arabic: النظام الآلي لكشف الشذوذ

Priority: High

🔴 **UNIQUE FEATURE - AI-Powered Real-Time Monitoring**

4.7.1 Indicator Anomaly Alerts

Route: /intelligence/anomaly-detection/indicators

- Statistical anomaly detection
- Threshold breach alerts
- Unusual pattern flagging
- Regime change identification

4.7.2 Transaction Anomalies

Route: /intelligence/anomaly-detection/transactions

- Unusual trade patterns
- Currency arbitrage detection
- Capital flight indicators
- Remittance flow irregularities
- Trade pattern deviations

4.7.3 Alert Management Dashboard

Route: /intelligence/anomaly-detection/alerts

- Real-time alert feed
- Severity classification
- Alert routing
- Investigation workflow
- False positive management

4.8 Natural Language Query Interface

Route: /intelligence/nl-query

English: Ask Questions in Plain Language

Arabic: اسأل بلغة عادية

Priority: High

🟢 **UNIQUE FEATURE - Conversational AI**

Features:

- Natural language question input
- AI-powered query understanding
- Intent classification
- Natural language answer generation
- Follow-up question support
- Context retention
- Multi-language support (EN/AR)
- Voice input
- Query history
- Saved queries

Example Queries:

- "What was the impact of the 2016 Aulaqitah offensive on food prices?"
- "Compare remittance flows in 2015 vs 2020"
- "What drivers fueled the most adjustable program?"
- "Which key commodities showed the most volatile price movements?"

4.9 Pattern Recognition Engine

Route: /intelligence/pattern-recognition

English: Economic Pattern Recognition

Arabic: التعرف على الأنماط الاقتصادية

Priority: Medium

4.9.1 Cycle Detection

Route: /intelligence/pattern-recognition/cycles

- Business cycle identification
- Seasonal patterns
- Conflict recurrency cycles
- Aid dependency cycles

4.9.2 Correlation Discovery

Route: /intelligence/pattern-recognition/correlations

- Automated correlation mining
- Lag structure analysis
- Causality testing (Granger)
- Spurious correlation filtering

4.9.3 Regime Identification

Route: /intelligence/pattern-recognition/regimes

- Structural break detection
- Regime switching models
- Hidden Markov models
- Stability analysis

4.10 Learning System

Route: /intelligence/learning-system

English: Continuous Learning & Model Improvement

Arabic: التعلم المستمر وتحسين النماذج

Priority: Medium

🔴 **UNIQUE FEATURE - Self-Improving AI**

4.10.1 Model Performance Tracking

Route: /intelligence/learning-system/performance-tracking

- Forecast accuracy metrics (RMSE/MAE/MAPE)
- Model comparison leaderboards
- Performance degradation detection
- Retraining triggers

4.10.2 Feedback Integration

Route: /intelligence/learning-system/feedback

- User feedback collection
- Expert validation input
- Outcome verification
- Model adjustment based on feedback

4.10.3 Auto-Retraining Pipeline

Route: /intelligence/learning-system/auto-retrain

- Scheduled retraining
- Trigger-based retraining
- Hyperparameter optimization
- Model versioning